The effects of zinc supplementation on wound healing and metabolic status in patients with diabetic foot ulcer: a randomized, double-blind, placebo-controlled trial

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ABSTRACT

This study was performed to determine the effects of zinc supplementation on wound healing and metabolic status in patients with diabetic foot ulcer. The current randomized, double-blind, placebo-controlled trial was conducted among 60 patients (aged 40-85 years old) with grade 3 diabetic foot ulcer. Participants were randomly divided into two groups (30 participants in each group) to take either 220 mg zinc sulfate supplements containing 50 mg elemental zinc or placebo daily for 12 weeks. After the 12-week intervention, compared with the placebo, zinc supplementation was associated with significant reductions in ulcer length (-1.5±0.7 vs. -0.9±1.2 cm, P=0.02) and width (-1.4±0.8 vs. -0.8±1.0 cm, P=0.02). In addition, changes in fasting plasma glucose (-40.5±71.0 vs. -3.9±48.5 mg/dL, P=0.02), serum insulin concentration (-8.0±15.4 vs. +1.1±10.3 μIU/mL, P=0.009), homeostasis model of assessment-estimated insulin resistance (-3.9±7.1 vs. +0.8±5.9, P=0.007), the quantitative insulin sensitivity check index (+0.01±0.03 vs. -0.002±0.02, P=0.04) and HbA1c (-0.5±0.8 vs. -0.1±0.5%, P=0.01) in the supplemented group were significantly different from the changes in these indicators in the placebo group. Additionally, significant increases in serum HDL-cholesterol (+4.1±4.3 vs. +1.1±5.1 mg/dL, P=0.01), plasma total antioxidant capacity (+91.7±213.9 vs. -111.9±188.7 mmol/L, P<0.01) and total glutathione (+68.1±140.8 vs. -35.0±136.1 μmol/L, P=0.006), and significant decreases in high sensitivity C-reactive protein (-20.4±24.6 vs. -6.8±21.3 μg/mL, P=0.02) and plasma malondialdehyde concentrations (-0.6±0.9 vs. -0.2±0.7 μmol/L, P=0.03) were seen following supplementation with zinc compared with the placebo. Zinc supplementation for 12 weeks among diabetic foot ulcer patients had beneficial effects on parameters of ulcer size and metabolic profiles.

Keywords: Zinc supplementation, wound healing, metabolic status, diabetic foot

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INTRODUCTION

The diabetic foot ulcer (DFU) is defined as manifestations of an invasion and multiplication of microorganisms in soft tissues or bone anywhere below the malleoli in a person with diabetes.[1] DFU may ultimately result in amputation, especially when wound infection or osteomyelitis are involved.[2] Its prevalence ranges from 4 to 10%, whereas, the lifetime risk for the development of a DFU in persons with diabetes ranges from 15 to as high as 25%.[3,4] Ischaemia, neuropathy and infection are the three pathological ingredients that result in DFU complications.[5] In addition, several studies have reported that hyperglycemia, insulin resistance, dyslipidemia, increased biomarkers of inflammation and oxidative stress play a significant role in the pathogenesis of DFU.[6,7]

Previous studies have shown that diabetes affects zinc homeostasis in many ways, although it is most probably the hyperglycemia and insulin resistance, rather than any primary lesions related to diabetes, which is responsible for the increased urinary loss and decrease in total body zinc.[8,9] In addition, zinc element plays an important role in the synthesis, storage of insulin, and altered pancreatic insulin content.[10] Few studies have reported that zinc levels were significantly lower in patients with DFU compared to those without DFU.[11] Furthermore, the favorable effects of zinc supplementation on markers of insulin metabolism, circulating biomarkers of inflammation and oxidative stress among patients without DFU were reported. We have previously found that 233 mg zinc gluconate (containing 30 mg zinc) supplementation for 6 weeks among women with gestational diabetes (GDM) had beneficial effects on insulin metabolism, lipid profiles, and biomarkers of inflammation and oxidative stress.[12,13] However, no significant change in fasting glucose, insulin and lipid concentrations was observed following supplementation with 240 mg zinc as zinc gluconate per day among patients with type 2 diabetes mellitus (T2DM) with normal zinc levels for 3 months.[14]

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Zinc intake may improve wound healing, insulin resistance, lipid profiles, biomarkers of inflammation and oxidative stress through its effect on enhancing the immune response,[8] decreased proliferative response of T and B lymphocytes [15] and the regulation of nuclear factor kappa-light-chain-enhancer of activated B cells (NF-κB) activation.[16] As there is evidence that zinc supplementation restores normal healing rates only in subjects who were zinc deficient and has anti-diabetic effect, we hypothesized that zinc supplementation might help patients with DFU to heal their wound faster, and have better metabolic profiles, and biomarkers of inflammation and oxidative stress. The objective of this study, therefore, was to evaluate the effects of zinc supplementation on wound healing and metabolic status in patients with DFU.

METHODS

Participants

This randomized double-blind placebo-controlled clinical trial, registered in the Iranian website for registration of clinical trials (http://www.irct.ir: IRCT201506215623N46), was done among 60 patients with grade 3 DFU according to "Wagner-Meggitt's" criteria aged 40-85 years who referred to the Shahid Beheshti Clinic in Kashan, Iran, between August 2015 and November 2015. This research was performed according to the principals of the Declaration of Helsinki and informed consent was taken from all patients. The trial was approved by the ethics committee of Kashan University of Medical Sciences (KUMS). Grade 3 DFU is defined as deep ulcer with abscess or osteomyelitis.[1] Pregnant and breastfed patients, participants who consumed zinc supplements during past 3 months, change in consuming medications throughout the study and patients with history of diseases which influence the development of DFU including chronic trauma, were excluded.

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Study design

Firstly, all subjects were matched for gender (males:21 and 9 females in each group), type and dosage of medications, duration of diabetes mellitus (DM), pre-treatment BMI (<25 and ≥25 kg/m^2) and age (<55 and ≥55 years). Individuals were then randomly divided into two groups to intake either 220 mg zinc sulfate supplements containing 50 mg elemental zinc or placebo daily for 12 weeks. Zinc supplements and placebos capsules were similar in shape and size, and were manufactured by Alhavi (Tehran, Iran). Randomization assignment was done using computer-generated random numbers as blindness by a trained staff at the clinic. In addition, all participants underwent a similar treatment protocol for the Diabetic Foot, based on the Infectious Diseases Society of America (IDSA).[17] Participants were requested not to change their routine physical activity and not to take any nutritional supplements during the 12-week trial. All patients completed 3-day food records and 3-physical activity records at baseline, weeks 3, 6 and 9 of the intervention and end-of-the trial. Daily macro- and micro-nutrient intakes were analyzed by nutritionist IV software (First Databank, San Bruno, CA). In the current study, physical activity was described as metabolic equivalents (METs) in hours per day.[18]

Treatment adherence

The use of zinc supplements and placebos during the study was checked by asking participants to return the medication containers. To increase compliance, all participants received brief daily cell phone reminders to intake the supplements. Compliance with the consumption of supplements and placebos was also checked by the measurement of serum zinc levels with enzymatic method.

Assessment of anthropometric measures

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Weight and height of participants were determined in an overnight fasting status using a standard scale (Seca, Hamburg, Germany) at baseline and after the 12-week intervention. BMI was calculated as weight in kg divided by height in meters squared.

Assessment of outcomes

In the current study, wound healing and markers of insulin metabolism were considered as the primary outcome measurements and lipid profiles, and biomarkers of inflammation and oxidative stress were considered as the secondary outcomes measurements.

Clinical assessment

Mean ulcer area was estimated as the product of longest measured length times longest perpendicular to length. Ulcer volume was estimated as the estimated area times the deepest ulcer depth. Depth ulcer was recorded as superficial or deep in addition to recording a measured depth. Infection was diagnosed if edema, erythema, discharge, regional lymph node enlargement, pain, or fever was present.[17] Participants were considered positive for infection if at least two of the above listed features were present. According to their location, ulcers were defined as plantar or nonplantar. Each foot was evaluated and graded according to the Wagner classification.[17]

Biochemical assessment

At baseline and after the 12-week intervention, 10 mL blood samples were obtained from each patient at Kashan reference laboratory in an early morning after an overnight fast. Then, the samples were stored at -80$^{\circ C}$ before analysis. A serum zinc concentration was evaluated using zinc kit (Hayyan, Tehran, Iran). Serum insulin concentrations were assessed using available ELISA kit (DiaMetra, Milano, Italy). The homeostasis model of assessment-insulin

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resistance (HOMA-IR), β-cell function (HOMA-B) and the quantitative insulin sensitivity check index (QUICKI) were determined according to the suggested formulas.[19] Hemoglobin A1c ($HbA1_C$) levels in the whole blood were quantified by Glycomat kit (BiocodeHycel, Massy, France). Enzymatic kits (Pars Azmun, Tehran, Iran) were used to quantify FPG, serum triglycerides, VLDL-, total-, LDL- and HDL-cholesterol concentrations. All inter- and intra-assay CVs for FPG and lipid levels were less than 5%. Erythrocyte sedimentation rate (ESR by Westergren method) was obtained at baseline and after the 12-week treatment. Serum high sensitivity C-reactive protein (hs-CRP) concentrations were assessed by commercial ELISA kit (LDN, Nordhorn, Germany). The plasma nitric oxide (NO) concentrations were assessed using Griess method.[20] Plasma total antioxidant capacity (TAC) concentrations by the method of ferric reducing antioxidant power developed by Benzie and Strain,[21] total glutathione (GSH) using the method of Beutler et al.[22] and malondialdehyde (MDA) concentrations by the thiobarbituric acid reactive substances spectrophotometric test [23] were evaluated. All inter- and intra-assay CVs for NO, TAC, GSH and MDA concentrations were less than 5%.

Sample size

To calculate the sample size, we used a parallel randomized clinical trial sample size formula where type one (α) and type two errors (β) were 0.05 and 0.20 (power=80%), respectively. Based on a previous study,[24] we considered 1.57 as standard deviation (SD) and 1.30 as the difference in mean (d) of HOMA-IR as the key variable. Therefore, we needed 25 participants in each group. Considering 5 dropouts in each group, the final sample size was determined to be 30 participants each group.

Statistical methods

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We applied the Kolmogrov-Smirnov test to determine the normal distribution of variables. The final analysis was carried out according to intention-to-treat (ITT) principle. Missing values were treated based on Last-Observation-Carried-Forward method (LOCF).[25] LOCF ignores whether the participant's condition was improving or deteriorating at the time of dropout but instead freezes outcomes at the value observed before dropout (i.e., last observation). To detect differences in anthropometric measures as well as in dietary intakes between the two groups, we applied independent samples Student's t test. To determine the effects of zinc supplementation on ulcer size, glycemic status, lipid concentrations, biomarkers of inflammation and oxidative stress, we used one-way repeated measures analysis of variance. To identify within-group differences (baseline and end-of-trial), we used paired-samples t-tests. Differences in proportions were evaluated by Chi square test or Fisher's exact tests. In addition, to identify within-group differences (baseline study and end-of-trial) of nominal variables, we used the non-parametric McNemar test. To control confounding variables including baseline values, age and baseline BMI, we used analysis of covariance (ANCOVA). P-values <0.05 were considered statistically significant. All statistical analyses were done by the use of the Statistical Package for Social Science version 18 (SPSS Inc., Chicago, Illinois, USA).

RESULTS

Among participants in the placebo group, 2 participants [withdrawn due to personal reasons (n=2)] were excluded **(Fig.1).** At the end, 58 participants with DUF [zinc (n=30) and placebo (n=28)] completed the trial. However, as the analysis was based on the ITT principle, all 60 patients (30 participants in each group) were included in the final analyses.

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Gender distribution, mean age, height, weight and BMI at baseline and end-of-trial, METs at baseline and end-of-trial, and insulin and metformin therapy were not statistically different between the two groups **(Table 1).**

Based on the 3-day dietary records obtained at baseline, end-of-trial and throughout the trial, we found no significant difference in mean dietary macro- and micro-nutrient intakes between the two groups **(Table 2).**

Zinc supplementation resulted in a significant rise in serum zinc levels (+12.7±4.7 vs. -3.5±4.0 mg/dL, P<0.001) compared with the placebo. After the 12-week intervention, compared with the placebo, zinc supplementation was associated with significant reductions in ulcer length (-1.5±0.7 vs. -0.9±1.2 cm, P=0.02) and width (-1.4±0.8 vs. -0.8±1.0 cm, P=0.02) **(Table 3).** In addition, changes in FPG (-40.5±71.0 vs. -3.9±48.5 mg/dL, P=0.02), serum insulin concentration (-8.0±15.4 vs. +1.1±10.3 µIU/mL, P=0.009), HOMA-IR (-3.9±7.1 vs. +0.8±5.9, P=0.007), HOMA-B (-15.0±35.7 vs. +1.4±19.2, P=0.03), QUICKI (+0.01±0.03 vs. -0.002±0.02, P=0.04) and HbA1c (-0.5±0.8 vs. -0.1±0.5%, P=0.01) in the supplemented group were significantly different from the changes in these indicators in the placebo group. Additionally, significant increases in serum HDL-cholesterol (+4.1±4.3 vs. +1.1±5.1 mg/dL, P=0.01), plasma TAC (+91.7±213.9 vs. -111.9±188.7 mmol/L, P<0.01) and GSH (+68.1±140.8 vs. -35.0±136.1 µmol/L, P=0.006), and significant decreases in total-/HDL-cholesterol ratio (-0.7±0.9 vs. -0.2±0.9, P=0.03), hs-CRP (-20.4±24.6 vs. -6.8±21.3 µg/mL, P=0.02), ESR (-34.6±28.5 vs. -13.3±28.4 mm/h, P=0.005) and plasma MDA concentrations (-0.6±0.9 vs. -0.2±0.7 µmol/L, P=0.03) were seen following supplementation with zinc compared with the placebo. There were no significant changes in other lipid profiles and biomarkers of inflammation between the two groups. However, we done analysis

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without ITT approach (excluding subjects who had withdraw), no significant change was seen in our findings.

When we adjusted the analysis for baseline values of biochemical parameters, age and baseline BMI, the difference in changes in ulcer depth (P=0.04) and NO (P<0.001) between the two groups were statistically significant, while difference in change in HOMA-B (P=0.07) was non-significant **(Table 4).** Other findings did not alter significantly after this adjustment.

DISCUSSION

In the current study, which to our knowledge is the first of its kind, we assessed effects of zinc supplementation on wound healing, glucose metabolism, lipid profiles, and biomarkers of inflammation and oxidative stress among patients with DFU. We found that taking zinc supplements for 12 weeks among patients with DFU had beneficial effects on parameters of ulcer size, glucose metabolism, HDL-, total-/HDL-cholesterol ratio, hs-CRP, ESR, plasma TAC, GSH and MDA levels; however, it did not affect other lipid profiles and biomarkers of inflammation.

We shown that zinc supplementation among patients with DFU for 12 weeks led to a significant improvement in parameters of wound healing compared with the placebo. It must be kept in mind that observed change in parameters of ulcer size in our study between the two groups was statistically significant; but it was not clinically significant. On the other hand, the percentage of those who was completely treated after intervention was not significantly different between the two groups. Long-term interventions and higher dosage of zinc might result in greater changes in parameters of ulcer size. An observational study was reported that

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mean serum zinc levels were significantly lower in diabetics with foot ulcers compared to those without foot ulcers.[11] In another study by Kheiralla et al.[8] was observed that 25 mg/day zinc supplementation for 1 month among patients with DFU may help wound healing by enhancing the immune response. Hyperglycemia and impaired insulin metabolism in diabetic patients disrupts the normal process of cytokine production,[26] which in turn leads to the progression of chronic wound.[27] In addition, several physiological factors such as decreased growth factor production,[28] macrophage function [29] and impaired cytokine production [30] help to wound-healing abnormalities in diabetic subjects. Chronic diseases especially diabetes are associated with decreased circulating levels zinc which plays an important role for immune system cells and many enzymes and the decreases of humoral and cell-mediated immune system. Zinc levels are reduced by many non-specific factors such as infections and trauma.[31] Zinc intake improve wound healing through enhancing the immune response,[8] up-regulation of metallothioneins [32] and zinc metalloenzymes like alkaline phosphatase, RNA and DNA polymerases, and matrix metalloproteinases (MMPs).[33]

This study demonstrated that zinc supplementation compared with the placebo in patients with DFU for 12 weeks had potential to decrease FPG, serum insulin, HOMA-IR, HOMA-B, HbA1c and total-/HDL-cholesterol ratio, and to increase QUICKI and HDL-cholesterol levels, but did not affect other lipid profiles. Our previous study shown that taking 30 mg/day zinc supplements for 6 weeks among GDM women resulted in significant decreases in FPG, serum insulin concentrations, HOMA-IR, HOMA-B, serum triglycerides, VLDL-cholesterol and a significant rise in QUICKI.[12] In addition, supplementation with 20 mg/day elemental zinc for 8 weeks in prepubertal obese children led to significant reductions in FPG, insulin and HOMA-IR, but did not influence lipid profiles.[34] A significant decrease in fasting blood sugar, triglycerides, VLDL-cholesterol levels, and a significant increase in HDL-cholesterol

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concentrations was observed following supplementation with 50 mg elemental zinc as zinc sulphate for 12 weeks among T2DM with microalbuminuria patients, but did not affect other lipid profiles.[35] However, no significant change in glucose and insulin concentrations was seen after zinc supplementation (240 mg/day of zinc as zinc gluconate) among patients with T2DM with normal zinc levels for 3 months.[14] Hyperglycemia and insulin resistance are associated with an acceleration of arteriosclerotic changes of the greater arteries resulting in decreased blood flow, further contributing to a limb threatening ischemic condition.[36] The diabetic dyslipidemia is one of several factors that contribute to developing macrovascular disease in diabetic patients.[37] Zinc intake may improve parameters of glucose homeostasis through the synthesis, storage, and secretion of monomeric insulin, as well as conversion to dimeric form for storage and secretion as crystalline insulin and participation in insulin action and carbohydrate metabolism.[10] It has been suggested that zinc intake may play an important role in enzymes involved in lipid metabolism and similar action to insulin.[38,39]

The current study indicated that compared with the placebo, zinc administration for 12 weeks among patients with DFU was associated with significant decreases of serum hs-CRP, ESR and plasma MDA concentrations, and significant increases of plasma TAC and GSH levels, but did not affect plasma NO concentrations. Supporting with our results, Kim and Ahn demonstrated that taking 30 mg/day supplemental zinc as zinc gluconate for 8 weeks among young obese women decreased hs-CRP levels.[40] In addition, Mazani et al.[41] observed that 100 mg/day zinc supplementation for 2 months among patients with hemodialysis was associated with a significant rise in TAC levels. Decreased circulating levels of MDA among diabetic patients have also reported following supplementation with 30 mg/day zinc for six months.[42] However, no significant effect in hs-CRP concentrations was seen after intake of 10 mg rosuvastatin for 4 months with or without 30 mg/day zinc in subjects with atherosclerosis.[43]

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Moreover, 233 mg zinc gluconate (containing 30 mg zinc) administration among GDM women for 6 weeks did not influence MDA levels.[13] In subjects with DFU, "adipovascular axis" expression in lower levels of adiponectin and higher levels of inflammatory cytokines could be linked to foot ulcers pathogenesis by inflammatory mechanisms.[44] Increased biomarkers of oxidative stress and free radicals play important roles in cell signaling and immune response during wound healing.[45] Therefore, the regulation of inflammatory and oxidative stress responses is an important factor in cutaneous wound healing. The beneficial effects of zinc intake on biomarkers of inflammation and oxidative stress may be mediated by the regulation of NF-κB activation through anti-inflammatory proteins A20, peroxisome proliferator activated receptor (PPAR)-α signaling pathway (16) and inhibiting the nicotinamide adenine dinucleotide phosphate-oxidase (NADPH-oxidase) enzyme.[46]

The current study had few limitations. Firstly, the sample size was small. Future studies with bigger sample size are needed to confirm the validity of our findings. Another limitation was that we could not assess gene expression related to insulin, inflammation and oxidative stress in patients with DFU.

Overall, taking zinc supplements for 12 weeks among patients with DFU had beneficial effects on parameters of ulcer size, glucose metabolism, HDL-, total-/HDL-cholesterol ratio, hs-CRP, ESR, plasma TAC, GSH and MDA levels; however, it did not affect other lipid profiles and biomarkers of inflammation.

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Authors' contributions

ZA contributed in conception, design, statistical analysis and drafting of the manuscript. MM-H, EB, RR, FB and HG contributed in data collection and manuscript drafting. All authors approved the final version for submission. ZA supervised the study.

Conflicts of interest

None declared.

Clinical trial registration number

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Table 1

General characteristics of study participants

	Placebo group (n=30)	Zinc group (n=30)	P Value*
Gender (%)			
Male	21 (70.0)	21 (70.0)	1.00†
Female	9 (30.0)	9 (30.0)	
Age (y)	60.0±10.0	58.3±8.6	0.46
Height (cm)	166.9±8.6	169.8±8.4	0.18
Weight at study baseline (kg)	72.5±12.8	74.9±12.1	0.45
Weight at end-of-trial (kg)	72.7±13.1	75.0±11.5	0.47
Weight change (kg)	0.2±1.9	0.1±2.2	0.83
BMI at study baseline (kg/m^2)	25.8±3.1	25.8±3.0	0.98
BMI at end-of-trial (kg/m^2)	25.9±3.3	25.9±3.1	0.98
BMI change (kg/m^2)	0.1±0.7	0.1±0.8	0.98
MET-h/day at study baseline	26.3±2.3	27.1±2.5	0.24
MET-h/day at end-of-trial	26.2±2.4	27.1±2.6	0.15
MET-h/day change	-0.1±0.9	0.01±1.0	0.43
Insulin therapy (%)	30 (100.0)	30 (100.0)	1.00†
Metformin therapy (%)	27 (90.0)	27 (90.0)	1.00†

Data are means± SDs.

* Obtained from independent t test. METs, metabolic equivalents.

† Obtained from Pearson Chi-square test.

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Table 2

Mean dietary intakes of study participants throughout the study

	Placebo group (n=30)	Zinc group (n=30)	P Value*
Energy (kcal/d)	2219±212	2183±220	0.52
Carbohydrates (g/d)	308.1±46.1	300.8±47.6	0.60
Protein (g/d)	81.1±15.6	80.0±15.6	0.77
Fat (g/d)	77.3±14.4	76.7±13.6	0.88
SFAs (g/d)	24.4±5.1	23.7±4.5	0.61
PUFAs (g/d)	23.6±5.7	24.0 ±6.2	0.80
MUFAs (g/d)	21.2±6.0	20.8±5.7	0.77
Cholesterol (mg/d)	201.5±111.8	185.0±100.1	0.53
TDF (g/d)	19.1±4.4	19.0±4.3	0.96
Zinc (mg/d)	9.9±2.7	9.7±2.4	0.67

Data are means± SDs. Values are mean dietary intakes of study participants at baseline, weeks 3, 6 and 9 of the intervention, and end-of-the trial.

* Obtained from independent t test.

MUFAs, monounsaturated fatty acids; PUFAs, polyunsaturated fatty acids; SFAs, saturated fatty acids; TDF, total dietary fiber.

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Table 3

Wound healing, metabolic profiles, and biomarkers of inflammation and oxidative stress at the baseline and after 12 weeks of the intervention in patients with diabetic foot ulcer

	Placebo group (n=30)				Zinc group (n=30)				P Value*[*]
	Baseline	End-of-trial	Change	P Value[*]	Baseline	End-of-trial	Change	P Value[*]	
Ulcer length (cm)	3.1±1.6	2.2±1.6	-0.9±1.2	<0.001	3.1±1.5	1.6±1.5	-1.5±0.7	<0.001	0.02
Ulcer width (cm)	2.7±1.4	1.9±1.0	-0.8±1.0	<0.001	2.9±1.2	1.5±1.3	-1.4±0.8	<0.001	0.02
Ulcer depth (cm)	1.3±0.6	1.0±0.9	-0.3±1.0	0.12	1.3±0.5	0.5±0.5	-0.8±0.6	<0.001	0.05
Ulcer symptoms									
Erythema (%)	30 (100.0)	11 (36.7)	-	<0.001†	30 (100.0)	9 (30.0)	-	<0.001†	0.58††
Discharge (%)	30 (100.0)	5 (16.7)	-	<0.001†	30 (100.0)	3 (10.0)	-	<0.001†	0.44††
Necrosis (%)	9 (30.0)	3 (10.0)	-	<0.001†	9 (30.0)	1 (3.3)	-	<0.001†	0.25††
Zinc (mg/dL)	77.6±12.9	74.0±13.3	-3.5±4.0	<0.001	76.4±4.5	89.1±6.7	12.7±4.7	<0.001	<0.001
FPG (mg/dL)	199.5±71.6	195.6±86.9	-3.9±48.5	0.66	187.6±73.6	147.1±50.0	-40.5±71.0	0.004	0.02
Insulin (μIU/mL)	18.5±9.7	19.7±10.7	1.1±10.3	0.54	21.1±13.2	13.1±8.2	-8.0±15.4	0.008	0.009
HOMA-IR	9.3±6.2	10.1±7.7	0.8±5.9	0.43	10.1±6.8	6.2±5.1	-3.9±7.1	0.006	0.007
HOMA-B	32.4±17.7	33.8±19.5	1.4±19.2	0.67	40.4±30.9	25.4±20.0	-15.0±35.7	0.02	0.03
QUICKI	0.29±0.02	0.28±0.02	-0.002±0.02	0.55	0.29±0.03	0.30±0.02	0.01±0.03	0.03	0.04
HbA1c (%)	7.9±0.7	7.8±0.6	-0.1±0.5	0.56	7.8±0.9	7.3±1.1	-0.5±0.8	0.003	0.01

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Triglycerides (mg/dL)	152.2±46.1	147.1±48.3	-5.1±33.4	0.40	127.7±78.4	119.8±62.6	-7.8±34.9	0.22	0.76
VLDL-cholesterol (mg/dL)	34.4±9.2	29.4±9.7	-1.0±6.7	0.40	25.5±15.7	24.0±12.5	-1.5±7.0	0.22	0.76
Total cholesterol (mg/dL)	163.4±45.9	160.4±49.1	-3.0±23.6	0.50	144.6±46.0	136.3±44.0	-8.3±26.4	0.09	0.41
LDL-cholesterol (mg/dL)	99.8±46.5	96.8±48.0	-3.0±24.1	0.49	85.9±35.2	75.1±34.7	-10.8±23.7	0.01	0.21
HDL-cholesterol (mg/dL)	33.1±6.5	34.3±6.6	1.1±5.1	0.23	33.1±7.0	37.2±8.1	4.1±4.3	<0.001	0.01
Total-/HDL-cholesterol ratio	5.1±1.5	4.9±1.8	-0.2±0.9	0.27	4.4±1.1	3.7±0.9	-0.7±0.9	<0.001	0.03
hs-CRP (μg/mL)	49.0±31.1	42.2±26.5	-6.8±21.3	0.09	53.3±21.3	33.9±27.1	-20.4±24.6	<0.001	0.02
ESR (mm/h)	63.5±37.6	50.2±29.7	-13.3±28.4	0.01	66.9±38.7	32.3±26.7	-34.6±28.5	<0.001	0.005
NO (μmol/L)	42.8±5.3	44.7±5.9	1.8±8.2	0.23	49.7±11.0	56.2±8.1	6.5±11.2	0.004	0.07
TAC (mmol/L)	1031.9±347.3	920.0±370.6	-111.9±188.7	0.003	862.6±201.3	954.3±168.7	91.7±213.9	0.02	<0.001
GSH (μmol/L)	550.3±101.3	515.3±91.8	-35.0±136.1	0.17	539.5±124.3	607.6±111.7	68.1±140.8	0.01	0.006
MDA (μmol/L)	3.5±0.8	3.3±0.6	-0.2±0.7	0.06	3.5±0.8	2.9±1.0	-0.6±0.9	<0.001	0.03

All values are means± SDs.

* P values represent paired-samples t-test.

** P values represent the time × group interaction (computed by analysis of the one-way repeated measures ANOVA).

† Obtained from McNemar test.

†† Obtained from Chi-square test.

ESR, erythrocyte sedimentation rate; FPG, fasting plasma glucose; GSH, total glutathione; HOMA-IR, homeostasis model of assessment-estimated insulin resistance; HOMA-B, homeostasis model of assessment-estimated B cell function; hs-CRP, high-sensitivity C-reactive protein; MDA, malondialdehyde; NO, nitric oxide; QUICKI, quantitative insulin sensitivity check index; TAC, total antioxidant capacity.

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Table 4

Adjusted changes in metabolic profile of the patients with diabetic foot ulcer

	Placebo group (n=30)	Zinc group (n=30)	P Value*
Ulcer length (cm)	-0.9±0.2	-1.5±0.2	0.02
Ulcer width (cm)	-0.9±0.1	-1.3±0.1	0.02
Ulcer depth (cm)	-0.3±0.1	-0.7±0.1	0.04
Zinc (mg/dL)	-3.5±0.8	12.7±0.8	<0.001
FPG (mg/dL)	-2.2±10.2	-42.1±10.2	0.008
Insulin (µIU/mL)	-0.1±1.7	-6.7±1.7	0.007
HOMA-IR	0.5±0.9	-3.5±0.9	0.005
HOMA-B	-2.2±3.5	-11.3±3.5	0.07
QUICKI	-0.003±0.004	0.01±0.004	0.004
HbA1c (%)	-0.04±0.1	-0.5±0.1	0.01
Triglycerides (mg/dL)	-2.1±5.6	-10.9±5.6	0.27
VLDL-cholesterol (mg/dL)	-0.4±1.1	-2.2±1.1	0.27
Total cholesterol (mg/dL)	-1.7±4.6	-9.6±4.6	0.24
LDL-cholesterol (mg/dL)	-1.8±4.4	-12.1±4.4	0.10
HDL-cholesterol (mg/dL)	1.1±0.9	4.1±0.9	0.01
Total-/HDL-cholesterol ratio	-0.1±0.2	-0.8±0.2	0.01
hs-CRP (µg/mL)	-7.4±3.9	-19.8±3.9	0.02
ESR (mm/h)	-16.1±5.3	-31.7±5.3	0.04
NO (µmol/L)	-1.1±1.3	9.4±1.3	<0.001
TAC (mmol/L)	-83.6±32.5	63.4±32.5	0.003

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GSH (μmol/L)	-30.9±18.5	64.0±18.5	0.001
MDA (μmol/L)	-0.2±0.1	-0.7±0.1	0.03

All values are means± SEs. Values are adjusted for baseline values, age and BMI at baseline.

[*] Obtained from ANCOVA.

ESR, erythrocyte sedimentation rate; FPG, fasting plasma glucose; GSH, total glutathione; HOMA-IR, homeostasis model of assessment-estimated insulin resistance; HOMA-B, homeostasis model of assessment-estimated B cell function; hs-CRP, high-sensitivity C-reactive protein; MDA, malondialdehyde; NO, nitric oxide; QUICKI, quantitative insulin sensitivity check index; TAC, total antioxidant capacity.

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Legend to figure:

Fig. 1. Summary of patient flow diagram.

Fig. 1

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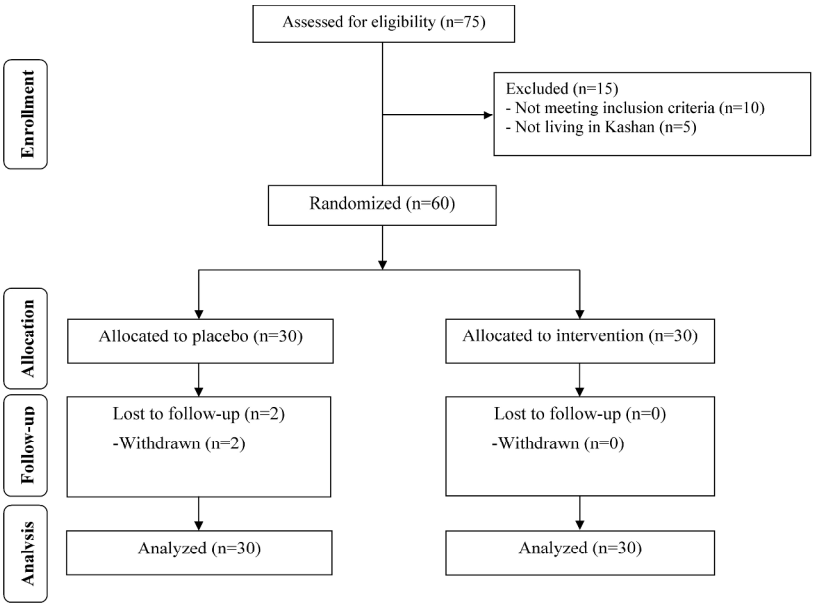

Fig. 1

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Figure

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